**QUESTION 1 [2+4 = 6 MARKS]**

After seeing the recent huge drop in the share market, you expect the market to recover and, thus, go up from current levels. You, therefore, went long futures on the ASX200 to speculate on your view. Specifically, you entered into ten September 2022 ASX200 futures contracts on the 6th June 2022 when the futures price was 7,175 points. On the 8th of June 2022, you increased your exposure by entering into eight additional September 2022 ASX200 futures contracts at the futures price of 7,065 points. Assume the initial margin required is $7,000 per contract and the maintenance margin is $3,500 per contract. Each index point is worth $50. You are provided with the following futures market prices:

| Date | Opening price | Last price | Settlement price |
|---|---|---|---|
| 6th June 2022 | 7,145 | 7,080 | 7,075 |
| 7th June 2022 | 7,073 | 7,050 | 7,045 |
| 8th June 2022 | 7,040 | 7,050 | 7,052 |

Please note: although you have taken two positions (one on June 6th, one on June 8th), they are taken on the same September 2022 contract. As a result, the cash flow from one position may offset the cash flow from the other. You need to look at the overall margin account when answering the questions below. Also, you need to account for margin calls: therefore, you need to compute end-of-the-day margin account balances after satisfying any required margin call.

**a)** What is your margin account balance at the end of 6th June 2022**? Please show all your detailed workings.**

**b)** Given the information above, will you have the right to withdraw money from the margin account at the close of the trading day on 8th June 2022? If not, why? If so, how much money at most would you be able to withdraw? **Please show all your detailed workings.**

**Answer:**

**a)**

To speculate on the increase of the market, you should long futures contracts.

For the first 10 contracts, initial margin = 7000*10=$70,000; maintenance margin = 3500*10=$35,000.

At the end of June 6th

- futures gain/loss = (7075-7175)*50*10=$-50,000
- Account balance = 70,000-50,000=20,000, which is lower than the maintenance margin. A margin call is required to top up the account to $70,000.
- Ending balance after margin call is met = $70,000 **2 MARKS**

b)

At the end of June 7th:

- Futures gain/loss = (7045-7075)*50*10=$-15,000
- Account balance = 70,000-15,000 = 55,000, which is above the maintenance margin.
- Ending balance = $55,000 **1 MARK**

At the end of June 8th:

- Initial margin required for the 8 contracts = 7,000*8 = $56,000
- Futures gain/loss from first 10 contracts = (7052-7045)*50*10=$3,500
- Futures gain/loss from the second 8 contracts = (7052-7065)*50*8=-$5,200

- Ending balance = 55,000+56,000+3,500-5,200 = **$109,300** which is higher than the maintenance margin of 3500*18=$63,000 but lower than the combined initial margins of $(70,000 + 56,000) = $126,000

  **2 MARKS**

➔ Since the margin account balance is not higher than the initial margin, money cannot be withdrawn from the margin account

**1 MARK**

**QUESTION 2 [2+2+1+2+3=10 marks]**

It is Friday, July 15th, 2022 and Marcell Jacobs, the head treasurer of OlympicGold plc., is looking at the futures quotes for silver reported below. The price quotes are in $ per ounce, while the contract size is 5,000. The contracts are traded on CME Globex. Assume Jacobs can trade at the last recorded price.

barchart.com/futures/quotes/SI*0/futures-prices

**SILVER PRICES** for Fri, Jul 15th, 2022 Alerts

Find the latest Silver prices and Silver futures quotes for all active contracts below.

Intraday | Main View | options quotes | flip

Latest futures price quotes as of Fri, Jul 15th, 2022.

| Contract | Last | Change | Open | High | Low | Previous | Volume | Open Int |
|---|---|---|---|---|---|---|---|---|
| SIY00 (Cash) | 18.6953 | +0.2718 | 18.4220 | 18.7140 | 18.2148 | 18.4235 | 4,606 | N/A |
| SIN22 (Jul '22) | 18.540 | +0.371 | 18.225 | 18.555 | 18.225 | 18.169 | 218 | 280 |
| SIQ22 (Aug '22) | 18.555 | +0.367 | 18.205 | 18.640 | 18.085 | 18.188 | 289 | 1,122 |
| SIU22 (Sep '22) | 18.640 | +0.415 | 18.335 | 18.695 | 18.095 | 18.225 | 51,468 | 121,935 |
| SIZ22 (Dec '22) | 18.800 | +0.414 | 18.495 | 18.835 | 18.250 | 18.386 | 1,438 | 22,433 |
| SIF23 (Jan '23) | 18.791 | +0.349 | N/A | 18.791 | 18.791 | 18.442 | 14 | 240 |
| SIH23 (Mar '23) | 18.895 | +0.342 | 18.635 | 18.905 | 18.635 | 18.553 | 100 | 2,563 |
| SIK23 (May '23) | 18.998 | +0.336 | N/A | 18.998 | 18.998 | 18.662 | 2 | 161 |
| SIN23 (Jul '23) | 19.121 | +0.328 | N/A | 19.121 | 19.121 | 18.793 | 1 | 139 |
| SIU23 (Sep '23) | 19.250 | +0.328 | N/A | 19.250 | 19.250 | 18.922 | 11 | 11 |
| SIZ23 (Dec '23) | 19.437 | +0.328 | N/A | 19.437 | 19.437 | 19.109 | 13 | 192 |
| SIF24 (Jan '24) | 19.472 | +0.328 | N/A | 19.472 | 19.472 | 19.144 | N/A | N/A |
| SIH24 (Mar '24) | 19.624 | +0.328 | N/A | 19.624 | 19.624 | 19.296 | N/A | N/A |
| SIK24 (May '24) | 19.694 | +0.328 | N/A | 19.694 | 19.694 | 19.366 | N/A | N/A |
| SIN24 (Jul '24) | 19.766 | +0.328 | N/A | 19.766 | 19.766 | 19.438 | N/A | 3 |
| SIZ24 (Dec '24) | 20.050 | +0.328 | N/A | 20.050 | 20.050 | 19.722 | N/A | 4 |
| SIN25 (Jul '25) | 20.213 | +0.328 | N/A | 20.213 | 20.213 | 19.885 | N/A | N/A |
| SIZ25 (Dec '25) | 20.325 | +0.328 | N/A | 20.325 | 20.325 | 19.997 | N/A | N/A |
| SIN26 (Jul '26) | 21.116 | +0.328 | N/A | 21.116 | 21.116 | 20.788 | N/A | 5 |

Jacobs is considering silver futures as their price has a positive correlation with the price of lightitan, a metal OlympicGold needs regularly in order to manufacture derailleurs for mountain bikes, but does not have listed futures written on. Following the company's policy, Jacobs is about to set up a hedging strategy aiming to minimize the price risk originating from a large purchase of lightitan. The latter purchase will be executed on September 15, 2022, and will be for one million ounces.

The research team has provided Jacobs with the following market statistics:

| | Summary Statistics of Price Changes – Annualized | | | |
|---|---|---|---|---|
| | Mean | Std Dev | Beta | Alpha |
| Lightitan Spot | 1.15 | 10.52 | 0.72 | 0.16 |
| Silver Spot | 0.22 | 9.60 | 0.55 | 0.035 |
| Silver Futures | 0.23 | 9.69 | 0.57 | 0.025 |
| | | | | |
| Note: Alpha and Beta are computed with respect to the S&P500 index | | | | |

| | Correlation Matrix of Price Changes – Annualized | | |
|---|---|---|---|
| | Lightitan Spot | Silver Spot | Silver Futures |
| Lightitan Spot | 1 | | |
| Silver Spot | 0.716 | 1 | |
| Silver Futures | 0.722 | 0.989 | 1 |

a) What position in silver futures should OlympicGold follow in order to hedge? Why?
Note: you are not required to work out the number of contracts in this part of the question.

The correlation between lightitan spot and Silver futures is positive, it needs to go LONG Silver futures

b) What Silver futures contract (i.e., what maturity) should Marcell Jacobs trade in? Explain your answer.

The December contract for the September purchase, to avoid September contract volatility
**1 MARK for correct contract, i.e., Dec. contract**
**1 MARK for correct reason, i.e., avoid delivery month volatility.**

c) How many contracts does Jacobs need to buy or sell in order to achieve the hedging objective outlined above? **Round to the nearest integer.**

h* = 0.722*10.52/9.69 = 0.783843

➔ N* = 0.783843*1,000,000/5,000 = 156.77 → 157 contracts

**1 MARK**

d) Given your answers in a), b) and c) above, summarize the complete hedging strategy for OlympicGold. Thus, please specify:
- the date the hedge is initiated,
- your position in the contract,

- the futures maturity,
- the number of contracts,
- the opening futures price
- how and when the hedge is closed out.

On July 15th, 2022, go long 157 December Silver Futures contracts at a price of $18.800

On September 15th, 2022, close out the Silver futures position by going short 157 December contracts

**1 MARK for transaction to open the hedge, including transaction date, contract maturity, number of contracts, futures price.**
**1 MARK for transaction to close the hedge, including transaction date, contract maturity, number of contracts.**

e) Can Marcell Jacobs determine as of today, 15/7/2022, the effective cost of lightitan for OlympicGold? If so, explain how. If not explain why not.
Ignore the impact of daily marking to market.

**No, the company cannot know at the time the hedge is set up (15/07/2022) what the effective price paid for the purchase of lightitan will be 1 MARK**

**The reason why they cannot know is basis risk, which is due (in this case) to asset mismatch and maturity mismatch. Specifically, they cannot know as of 15/7, at what futures price (say, $F_T$) they will close out the position on September 15th. Plus, they cannot know at what spot price (say, $S_T$) they will purchase lightitan on September 15th. Because of basis risk $F_T$ and $S_T$ will not necessarily be the same. 2 MARKS**

**QUESTION 3 [2+3=5 MARKS]**

At the end of trading on May 19, 2025, WTI crude oil settles at $80 per barrel in the spot market and at $86 per barrel on the 1-year futures contract. The risk-free interest rate is 5% per annum, continuously compounded and the cost of storing a barrel of oil for one year is $3, payable at the end of the year.

a) What convenience yield, if any, is the market attributing to WTI oil in % per year? Keep 2 decimal places for your final answer.

b) Given the information above, is there an exploitable arbitrage opportunity? Explain. You do not need to do any calculations in answering this question.

a) The present value of the storage costs per barrel is $3e^{-0.05\times1} = 2.854$. The one-year futures price based on the cost-of-carry is $(80+2.854)e^{0.05\times1} = 87.10$.

The convenience yield is the rate y that solves the following equation

86 *exp(y) = 87.10

➔ y = ln(87.10/86) = 1.27% p.a.

b) An actual futures price of $86 per barrel is below the no-arbitrage price and may suggest the existence of an arbitrage opportunity.

However, since light crude oil is a consumption commodity, market participants may not take advantage of such an opportunity. So, the low futures price is not necessarily an exploitable arbitrage; it can be interpreted as reflecting a positive convenience yield of about 1.27% p.a.

**QUESTION 4 [2+2+3=7 MARKS]** You may ignore this question as it was related to a case study that we did not cover this semester.

One of the FT articles covered in class describes investors using dispersion trades in response to AI-related stock-market shocks. The article explains that some investors are buying exposure to volatility in individual stocks while selling exposure to volatility in the S&P 500 index.

Based on the article and the class discussion, answer the following questions. Answer conceptually. No numerical calculations are required.

a) Explain what it means, in options-market terms, to "buy single-stock volatility". Why can this be different from simply betting that a particular stock price will rise?

b) Explain why selling index volatility may be profitable if the S&P 500 remains relatively stable. What risk does the investor take by selling index volatility?

c) The article notes that dispersion trades can be vulnerable if stocks begin to fall together. Explain why a market-wide sell-off can hurt a dispersion trade.

a) "buying single-stock volatility" means taking an options position that benefits from large movements in individual stocks, not necessarily only from price increases. A long option position has a premium cost but gives convex exposure: large favourable movements can generate large payoffs, while the option holder's downside is limited to the premium. Set 4 covers the distinction between calls and puts, the right-versus-obligation nature of options, and the fact that option payoffs are non-linear. **[2 marks]**

b) Selling index volatility means taking the opposite side of index-option protection. The investor receives option premium and benefits if the index remains calm, but bears losses if index volatility rises sharply or the index moves substantially. This fits the article's description of investors buying single-stock volatility while selling index volatility because individual companies are moving sharply even though the S&P 500 has stayed within a narrow range. **[2 marks]**

c) The key issue is correlation. A dispersion trade benefits when individual stocks move differently from one another and the index remains comparatively stable. If a broader shock causes stocks to

fall together, index volatility rises and the short index-volatility leg can lose money. The article explicitly warns that dispersion trades can come unstuck if a broader setback causes stocks to fall in unison. **[3 marks]**

**Question 5 (10 marks)**

A non-dividend paying stock has a current price of \$57. Its volatility is 10% p.a. and the continuously compounded risk-free rate is 7% p.a. for all maturities. An American put option on the stock has a strike price of \$62 and an expiry of 1 year. You need to value this option using a three-period binomial model. Show all your working.

a) Draw a binomial tree that displays the stock price and corresponding option value at every node over the life of this American put option. Keep at least 4 decimal places for $u$ and $d$ (the up and down factors in the binomial model), and keep at least 2 decimal places for the stock prices over the life of the option. **(7 marks)**

b) Following part (a), assume the option is currently priced at \$4. Identify any arbitrage opportunities that arise, and show your calculations along with the steps required to lock in a risk-free profit. (You are not required to present a table of cash flows in your answer.) **(3 marks)**

Solution:
See Excel Spreadsheet.

**Question 6 (12 marks)**
**Answer all parts of the question below.**

i. The current price of a non-dividend-paying stock is $60. The stock has an expected continuously compounded return of 12% per annum and a volatility of 25% per annum. Keep 4 decimal places for your answer in this question.
a) Construct a 95% confidence interval for the stock price in 9 months. Note: A 95% confidence interval is obtained by taking the mean plus or minus 1.96 standard deviations. **(3 marks)**
b) Calculate the expected value and variance of the stock price in 9 months. **(2 marks)**

ii. An index currently stands at 3,150. European call and put options on this index, both with a strike price of 2,950 and time to maturity of ten months are trading at 198.00 and 85.50, respectively. The risk-free rate is 5% p.a. with continuous compounding.
a) Calculate the implied dividend yield with continuous compounding that is consistent with the observed option prices. Show all your working. Express your final answer as a percentage, rounded to 2 decimal places. **(2 marks)**
b) If the actual dividend yield on the index is 14% p.a. with continuous compounding, describe the arbitrage opportunity that exists. Provide a table showing your cash flow calculations. Show all your working. Keep 2 decimal places for your final answer. **(5 marks)**

**Solution:**
i.

a) $\ln S_T \sim \phi\left[\ln 38 + \left(0.16 - \frac{0.35^2}{2}\right) * \frac{9}{12}, \left(0.35\sqrt{\frac{9}{12}}\right)^2\right]$

$\ln S_T \sim \phi[3.686961,\ \ 0.06125]$

95% confidence interval

$$3.686961 - 1.96 \times \sqrt{0.06125} < \ln S_T < 3.686961 + 1.96 \times \sqrt{0.06125}$$
$$24.5788 < S_T < 64.8474$$

$b)\ E(S_T) = S_0 e^{\mu T}$
$var(S_T) = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right)$
$\mathrm{E}[S_T] = 40.3498,\ \mathrm{Var}(S_T) = 662.7662$

ii.

a) $c + Ke^{-rT} = p + S_0 e^{-qT}$

$$198 + 2950e^{-0.05\times\frac{10}{12}} = 85.5 + 3150e^{-\frac{10}{12}q}$$
q=8.19%

b)
if q=12%, then c+K*e^(-rT)>p+S0*e^(-qT)

So we need to sell call, sell bond/lend K*e^(-rT), buy put, buy e^(-qT) shares with dividends being reinvested back into the shares

| | 0 | T | |
|---|---|---|---|
| | | ST>K | ST<=K |
| Sell Call | c | K-ST | 0 |
| Sell Bond/Lend | K*e^(-rT) | -K | -K |
| Buy Put | -p | 0 | K-ST |
| Buy e^(-qT) Shares with dividend reinvestment | -S0*e^(-qT) | ST | ST |
| **Net** | **-p-S0*e^(-qT)+c+K*e^(-rT)** | **0** | **0** |
| | | | |
| **Profit** | **138.98** | | |

**Note: Students must explicitly state that dividends are reinvested. If they fail to specify this, deduct 1 mark, even if the final calculation is correct.**

**Question 7 (10 marks)**

You have been employed by an Australian option dealer to keep an eye on its exposure to currency options. One area of concern that you have identified is the dealer's exposure to USD. Currently, the dealer is ensuring that its portfolio of short 10-month call options on USD is kept delta neutral by taking a position in 500,000 USD. Rather than using a position in USD, you are advocating a strategy to take a position in 10-month futures contracts on USD to achieve delta neutrality. Each futures contract is on 100,000 USD. Further, you have identified that the risk-free rates are 5% p.a. in Australia, and 6% p.a. in the US. Both rates are continuously compounded. Provide your final answers to four decimal places.

a) If the dealer were to take your advice, how could 10-month futures contracts on USD be used to make the dealer's position in currency options delta neutral? In your answer, calculate the number of futures contracts required and specify whether the position should be long or short. Assume that fractional futures contracts can be traded. (Hint: Australian dollar is the domestic currency, and USD is the foreign currency.) **(5 marks)**

b) If the dealer were to use 10-month forward contract instead of futures contract on USD, how can the dealer achieve delta neutrality? Note: Since forward contracts are tailored, you do not need to specify the number of contracts. Instead, state the USD amount to be covered by the forward contract. **(2 marks)**

c) Is your delta calculated in a) and b) the same? Provide an economic intuition. **(3 marks)**

**Solution:**

a) Step 1: Determine the position taken in the spot market.

Given a short call position, the option dealer must have taken a long position in USD. (1 mark)

Step 2: Calculate the delta of a futures contract on foreign currency.

The delta is given by the formula:

$\Delta_{\text{futures}} = e^{(r-r_f)T} = 0.9917013$ (1 mark)

This delta of 0.9917013 implies that for every 1 USD of futures exposure, the investor must hold 0.9917013 USD in the spot market to maintain the same risk profile.

Step 3: Determine the appropriate position in futures.

In this case, we are not using spot to hedge futures. Instead, we are using futures to replace the spot position in order to hedge the risk exposure in the options market. Therefore, the position in futures should mirror the position in the spot market:

- Long futures to replicate a long spot position.
  This ensures that movements in the USD price affect both positions similarly. (1 mark)

Step 4: Determine the number of futures contracts required.

As the delta of futures indicates the amount of spot exposure per unit of futures, the number of USD in futures contracts needed is

$500{,}000/0.9917013 = 504{,}184.0716$ (2 marks)

$\frac{504{,}184.0716}{100{,}000} = 5.04$ contracts

Final Answer: Long 5.0418 contracts

b) If forwards were to be used, then

$\Delta_{\text{forward}} = e^{-r_f T} = 0.9512294$

500,000/0.9512294=525,635.5482

Long forwards on 525,635.5482 USD

c) The delta of a long forward contract on currency is $e^{-r_f T}$, while the delta of a long futures contract is $e^{(r-r_f)T}$. In other words, $\Delta_{\text{futures}} = \Delta_{\text{forward}} * e^{rT}$. This is because the futures contract is marked to market daily and any gain or loss made on futures is immediately reflected in the margin account, while the value of the forward contract can only be realised at maturity.
This can be reflected in the formula below,

$$\text{Value of forward on currency} = (F - K)e^{-rT} = (S_0 e^{r-r_f T} - K)e^{-rT} = S_0 e^{-r_f T} - Ke^{-rT}$$

$$\text{Value of futures on currency} = F - K = S_0 e^{r-r_f T} - K$$

**Question 8 (10 marks)**

i. Suppose that the 6-month, 12-month, 18-month, and 24-month zero rates are 6.5% p.a., 7.2% p.a., 7.6% p.a., and 8.1% p.a. with continuous compounding. What is the two-year par yield with semi-annual compounding? Express your final answer in percentage and keep 2 decimal places. **(2 marks)**

ii. Assume all bonds in this question have a face value of $100. The prices of one-year and two-year zero-coupon bonds are 94.0 and 89.0, respectively. A three-year bond that pays annual coupons of $5 currently sells for $94.84. A four-year bond that pays annual coupons of $6 currently sells for $97.12. Calculate the three-year and four-year zero rates with continuous compounding. Express your final answers in percentage and keep 2 decimal places. **(3 marks)**

iii. Companies X and Y have been offered the following borrowing rates p.a. on a $10 million 3-year loan:

| | Fixed Rate (p.a.) | Floating Rate (p.a.) |
|---|---|---|
| Company X | 6.2% | SOFR |
| Company Y | 6.9% | SOFR |

Company X requires a floating-rate loan; company Y requires a fixed-rate loan. Design a swap that will net a bank, acting as intermediary, 0.1% p.a. and X shares 60% of the remaining gains and Y shares 40% of the remaining gains. Draw a swap diagram and show all your working. **(5 marks**)

**Solution:**

i. $$Ce^{-0.065\times0.5} + Ce^{-0.072\times1.0} + Ce^{-0.076\times1.5} + (100 + C)e^{-0.081\times2.0} = 100$$

C=4.1073

8.21% is the par yield.

ii. $94 * e^{R_1*1} = 100$ R1=0.061875

$89 * e^{R_2*2} = 100$ R2=0.058267

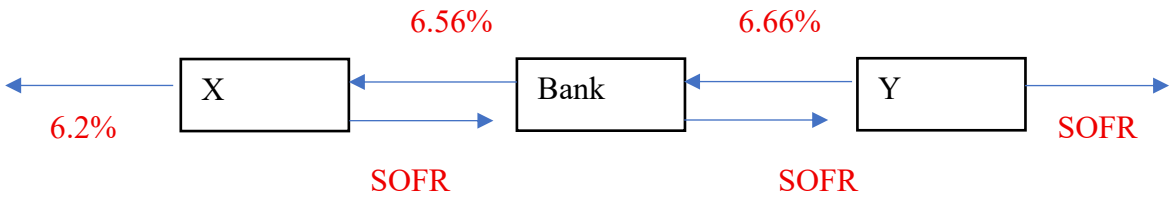
$$5e^{-0.061875\times1} + 5e^{-0.058267\times2} + 105e^{-R_3\times3} = 94.84$$

R3=6.77%

$$6e^{-0.061875\times1} + 6e^{-0.058267\times2} + 6e^{-0.04825\times3} + 106e^{-R_4\times4} = 97.12$$

R4=6.65%

iii. The spread between the interest rates offered to X and Y is 0.7% p.a. on fixed-rate loans and 0% p.a. on floating-rate loans. This means that the total benefit from the swap to all parties is 0.7%. Of this, 0.1% p.a. goes to the bank, 0.36% to Company X, and 0.24% to Company Y. As a result, Company X can borrow at a floating-rate return of SOFR-0.36%, while Company Y can borrow at a fixed-rate rate of 6.9%-0.24%=6.66% p.a. (2 marks) The required swap is illustrated below (3 marks).


6.56%
6.66%
X
Bank
Y
6.2%
SOFR
SOFR
SOFR